Asad Razzaq, PhD

Specially-Appointed Researcher, NISAR Lab, Kyoto University of Advanced Science, Kyoto, Japan
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Research Interests

Swarm robotics · evolutionary robotics · multi-robot learning and coordination · heterogeneous robot teams · neural network controllers · knowledge distillation · robotic olfaction · AI for human-robot and medical systems

Education

Ph.D. in Engineering — University of Toyama, Japan
Department of Mechanical and Intellectual Engineering — Oct 2022 – Sept 2025

- Thesis: *Evolutionary Design of Collective Behavior for Heterogeneous Swarm Robotic Systems*. Advisor: Prof. Toshiyuki Yasuda.
- Graduated as **Vice Valedictorian**. Funded by a JST SPRING doctoral fellowship.

Research Student — University of Toyama, Japan
Control Systems Engineering Laboratory — Apr 2022 – Oct 2022

M.S. in Computer Science — Government College University Faisalabad, Pakistan
Thesis: Impacts of Blockchain in Governance with Respect to Finance — Aug 2017 – Mar 2020

B.S. in Computer Science — Government College University Faisalabad, Pakistan
Final project: Urdu Steamer — Oct 2013 – Aug 2017

B.Ed. (Bachelor of Education) — Pakistan
Teaching qualification

Research and Professional Experience

Specially-Appointed Researcher — Kyoto University of Advanced Science, Japan
NISAR Lab, Department of Mechanical and Electrical Systems — Dec 2025 – present

- AI for multi-robot and human-machine systems: medical robotics, haptics, soft manipulators and actuators.
- Machine learning for multi-robot systems and autonomous exploration.

AI Systems Engineering Intern — Souya Corporation, Ishikawa, Japan
Deep learning and data analytics pipelines for enterprise clients — Sept 2025

Visiting Ph.D. Student — IRIDIA, Université libre de Bruxelles, Belgium
Supervised by Prof. Marco Dorigo — Mar 2025

- Blockchain for decentralized coordination, trust and data integrity in robot swarms.

Research Assistant — University of Toyama, Japan
Control Systems Engineering Laboratory — Oct 2022 – Sept 2025

- Evolutionary design of collective behavior in heterogeneous swarm robotic systems.
- Evolved neural controllers together with team composition for cooperative transport and collective exploration.

IT Executive — Approsoft Web Development and Design, Faisalabad, Pakistan
Full-cycle web development and client software — Apr 2020 – Mar 2022

Publications

Journal articles

[J1] **A. Razzaq**, T. Yasuda. Evolutionary design of group organization and controllers in heterogeneous robotic swarms. *Artificial Life and Robotics*, vol. 31, pp. 294–301, 2026. doi:10.1007/s10015-025-01083-5

[J2] **A. Razzaq**, T. Hayakawa, T. Yasuda. Evolutionary design of cooperative transport behavior for a heterogeneous robotic swarm. *Journal of Robotics and Mechatronics*, vol. 35, no. 4, pp. 1007–1015, 2023. doi:10.20965/jrm.2023.p1007

[J3] **A. Razzaq**, M. M. Khan, R. Talib, A. D. Butt, N. Hanif, S. Afzal, M. R. Raouf. Use of blockchain in governance: a systematic review of the literature. *International Journal of Advanced Computer Science and Applications*, vol. 10, no. 5, 2019. doi:10.14569/IJACSA.2019.0100585

International conference papers (peer-reviewed)

[C1] **A. Razzaq**, T. Yasuda. Knowledge distillation for developing versatile controllers of robotic swarms. In *Swarm Intelligence (ANTS 2026)*, Lecture Notes in Computer Science, vol. 16515, Springer, 2026. Darmstadt, Germany.

[C2] **A. Razzaq**, S. Wakayama, M. Manzoor, T. Yasuda. Evolutionary approach for optimization of perceptual abilities and group composition in heterogeneous robotic swarms. *SWARM 2025*, Germany, Sept. 2025.

[C3] **A. Razzaq**, T. Yasuda. Investigating swarm behavior through group compositions with perceptual variability in evolutionary swarm robotics system. *SICE FES 2025*, Thailand, Sept. 2025.

[C4] **A. Razzaq**, T. Yasuda. Evolutionary automatic design of heterogeneous robotic swarm composition for cooperative transport behavior. *SWARM 2024*, Kyoto, Japan, Sept. 2024.

[C5] **A. Razzaq**, T. Yasuda. Neural controller optimization for evolutionary heterogeneous swarm exploration. *SICE FES 2024*, Kochi, Japan, Aug. 2024.

[C6] **A. Razzaq**, T. Hayakawa, T. Yasuda. Investigating the influence of individual robot perception capability on swarm behavior in an evolutionary swarm robotic system. *AROB-ISBC-SWARM 2024*, pp. 1441–1446, Oita, Japan, Jan. 2024.

[C7] **A. Razzaq**, M. Takizawa, T. Hayakawa, T. Yasuda. Evolutionary acquisition of collective exploration behavior in heterogeneous SRS: investigation on neural controller configuration. *AROB-ISBC-SWARM 2023*, pp. 1424–1429, Oita, Japan, Jan. 2023.

[C8] M. A. Mushtaq, A. Sultan, **A. Razzaq**, Ehsaan-ur-Rehman, M. Inaam-ur-Rehman. New cryptographic algorithm based on CFA. *Proc. 4th International Conference on Big Data and Computing (ICBDC 2019)*, pp. 203–208, ACM, Guangzhou, China, May 2019. doi:10.1145/3335484.3335539

Poster

[P1] **A. Razzaq**, T. Yasuda. Evolutionary design of artificial swarms as sustainable systems. *Toyama Academic GALA 2024*, University of Toyama, Japan.

Theses

[T1] **A. Razzaq**. *Evolutionary Design of Collective Behavior for Heterogeneous Swarm Robotic Systems*. Ph.D. dissertation, University of Toyama, 2025.

[T2] **A. Razzaq**. *Impacts of Blockchain in Governance with Respect to Finance*. M.S. thesis, Government College University Faisalabad, 2020.

Awards and Funding

- **PhD Vice Valedictorian**, University of Toyama: second-highest-ranking doctoral graduate of the year (2025).
- **JST SPRING Doctoral Fellowship**, Toyama University Next Generation Researchers Challenging Research Program, Grant JPMJSP2145 (2023 – 2025).
- **SWARM 2025 Travel Grant**, to present at SWARM 2025 in Germany (2025).
- **Outstanding Student Award**, Society of Instrument and Control Engineers (SICE), Hokuriku Branch, Japan (2024).
- **Prime Minister's Laptop Scheme**, merit-based award, Government of Pakistan (2019).

Teaching and Mentoring

- B.Ed. (Bachelor of Education) teaching qualification.
- Guided undergraduate students in research and experimental methods, University of Toyama (2022 – 2025).
- Talks on careers, graduate study and scholarships for Pakistani student communities (2022 – present).
- **Courses I can teach.** Undergraduate: Programming Fundamentals (C/C++, Python), Object-Oriented Programming, Data Structures and Algorithms, Artificial Intelligence, Machine Learning, Introduction to Robotics. Graduate: Evolutionary Computation, Swarm Intelligence and Multi-Robot Systems, Deep Learning, Robot Learning.

Skills

Robotics and simulation	ROS / ROS 2, Gazebo, Isaac Sim, Isaac Lab, MuJoCo, Genesis, Drake, V-REP, Unity
AI and programming	Python, C/C++, C#, MATLAB, JavaScript, LaTeX; PyTorch, TensorFlow, CUDA, OpenCV, Git
Medical AI	MONAI, OpenSlide
Hardware	Raspberry Pi, Arduino, BeagleBone, UDOO, Pixhawk / ArduPilot, DJI Mavic, Parrot Bebop; Eagle, Altium Designer
Languages	English (proficient, C1), Japanese (basic, N5), Urdu (native)

References

Available on request.